Lecture 4

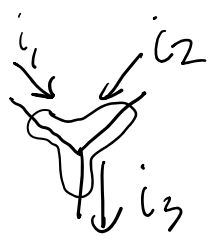
<u>Circuit Analysis:</u> Need to know voltages and currents throughout a circuit → need "tools" to do this

Kirchhoffs Laws

1.) <u>Kirchhoffs current Law (KCL)</u> → Conservation of charge.

States: the sum of currents entering a node is equal to the sum of currents leaving the node.

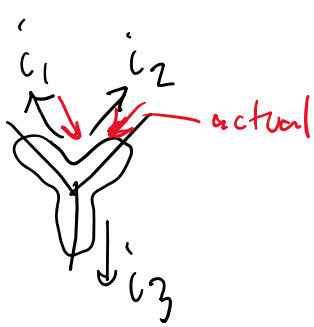
node = bubbles



* Assume currents leaving a node are Positive and entering is negative

$$+i_3 - i_1 - i_2 = 0$$

$$\boxed{i_3 = i_1 + i_2}$$

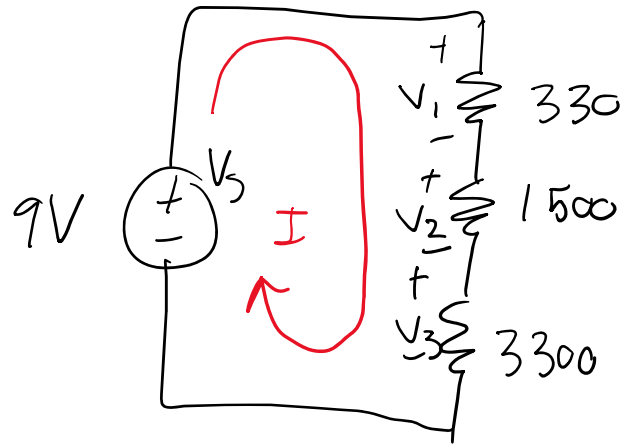


$$i_3 + i_1 + i_2 = 0$$

$$i_3 = -(i_1 + i_2)$$

Kirchhoff's Voltage Law (KVL)

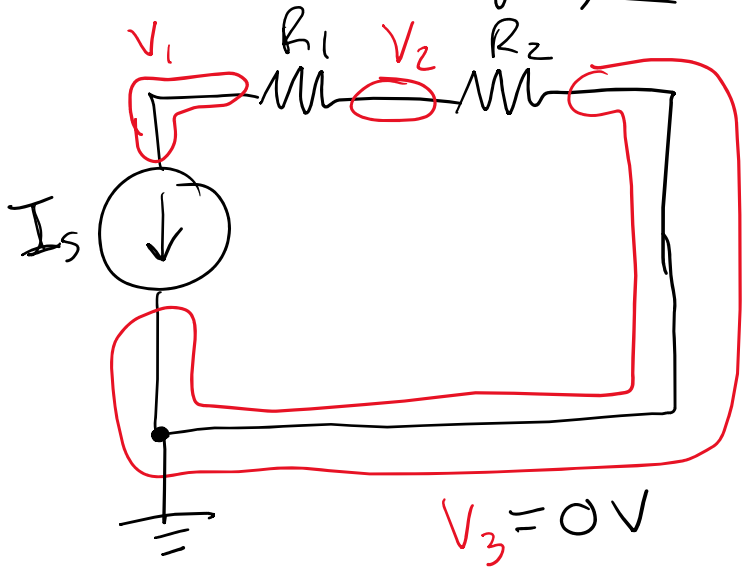
: the sum of all voltages around a Loop (mesh) is zero.



$$V_1 + V_2 + V_3 = V_s$$

$$-V_s + V_1 + V_2 + V_3 = 0$$

KCL (Nodal Analysis)



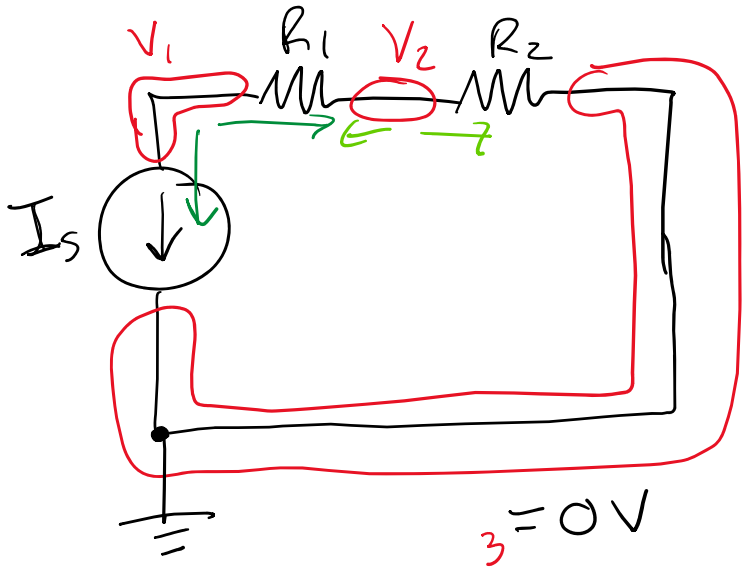
Step 1: Identify the nodes → each node corresponds to a voltage level in the circuit

Step 2: Assign a reference node.

* a reference is the "starting" point which has zero volts (0V).

* A good pratice is to make the node below the ~~the source~~ the Reference (common, ground, neutral)

main



Step 3: Make Eq. and Solve.

* Assume current always leaves the node.

* the node var. at, minus the adjacent node, divided by the element

The Trick: start at V_1 → make eq. for each node.

node 1

$$\frac{V_1 - V_2}{R_1} + I_s = 0$$

$$I_{thru\ R_1} + I_s = 0$$

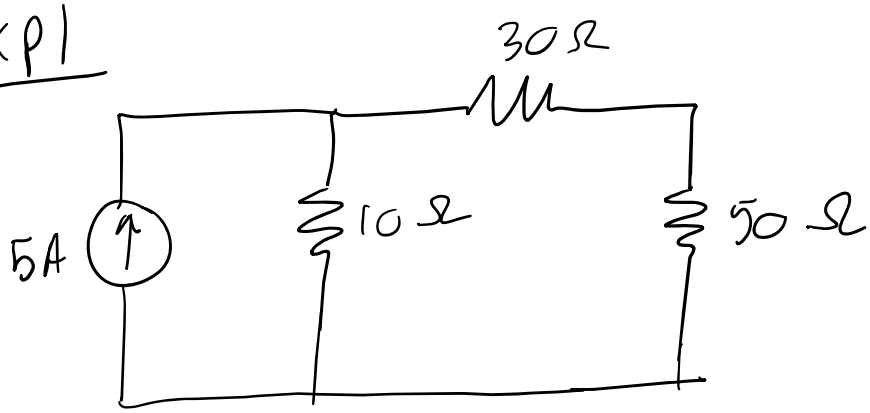
Node 2

$$\frac{V_2 - V_1}{R_1} + \frac{V_2 - V_3}{R_2} = 0 \quad ; V_3 = 0$$

$$\frac{V_2 - V_1}{R_1} + \frac{V_2}{R_2} = 0$$

Step 4: Solve through substitution.

Exp1



Step 1: Find nodes

Step 2: Assign Reference

Step 3: Make equations

Step 4: Solve